

[ENTER COMPANY NAME]

QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Post-Market Surveillance (PMS) Program

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Current Version / Revision	00
Effective Date	[Date of Final Approval]
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Applicable Regulatory Standards	ISO 13485:2016 Clause 8.2.1, 8.4, & 8.5 EU MDR 2017/745 Articles 83, 84, 85, & 86 FDA 21 CFR Part 822 (Postmarket Surveillance)

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1.0 Purpose

This Standard Operating Procedure (SOP) defines the system requirements for a continuous, proactive, and systematic process to collect, archive, and analyze real-world clinical performance data from distributed medical devices. The objective is to monitor the device's ongoing risk-benefit profile, identify unpredicted safety hazards, and provide input for clinical evaluation reviews, risk files, and design iterations.

2.0 Scope

This procedure is mandatory for all commercialized medical devices bearing corporate CE certification, FDA clearance/approval, or national licenses. It covers the entire lifecycle from commercial launch until the formal obsolescence of the last device profile in the market.

3.0 Responsibility and Authority

3.1 PMS Cross-Functional Team

A core compliance group comprised of members from Regulatory Affairs, Quality Assurance, Clinical Operations, and Engineering. The group meets regularly to execute PMS plan assessments.

3.2 Regulatory Affairs Manager

Holds organizational authority to draft specific product PMS Plans, compile annual trend profiles, and submit Periodic Safety Update Reports (PSUR) or Post-Market Surveillance Reports (PMSR) to Notified Bodies and Competent Authorities.

3.3 Clinical Affairs Specialist

Responsible for drafting Post-Market Clinical Follow-Up (PMCF) plans, designing active clinical surveys, and analyzing real-world clinical registries.

4.0 Procedure

4.1 PMS Planning Phase

Before launching a commercial product, a device-specific PMS Plan must be developed. The plan must detail the proactive and reactive methods to collect real-world performance metrics. These methods include:

- **Reactive Sources:** Customer complaints, adverse event logs, warranty returns, component failure metrics, and litigation records.
- **Proactive Sources:** Focus groups, specialized user surveys, scientific literature sweeps, clinical registry audits, and formal PMCF studies.

4.2 Data Collection and Literature Screening Matrix

On a defined semi-annual or annual frequency, the RA Specialist must perform targeted literature searches in major databases (such as PubMed, Embase, Cochrane) using specified search strings. This search captures data on the company's devices as well as equivalent or similar technologies to monitor state-of-the-art clinical benchmarks.

Output Document	Device Risk Classification	Compilation Frequency	Submission / Target Panel
Post-Market Surveillance Report (PMSR)	Class I (EU) / Class I (FDA)	Every 3 Years (or as required)	Maintained in the Technical File; made available to the Notified Body upon request.
Periodic Safety Update Report (PSUR)	Class IIa / Class IIb (EU)	At least every 2 Years	Submitted via electronic portal or uploaded directly into EUDAMED/ Notified Body vaults.
Periodic Safety Update Report (PSUR)	Class III / Implantable (EU)	Annually	Direct annual submission to the Notified Body and uploaded to EUDAMED.

4.3 Data Analysis Metrics and Trend Tracking

Proactive and reactive data must be evaluated to identify statistically significant trends. Data analysis checks whether the frequency or severity of a known risk has increased, or if a completely new hazard has been identified. Statistical parameters (such as a u-test or Poisson distribution calculations) are utilized to identify trend anomalies against established baseline metrics.

CRITICAL INTEGRATION MAPPING: If a PMS trend analysis identifies an unacceptable increase in risk frequency, the file must be escalated to the Risk Management Team to update the FMEA and immediately evaluate whether the CAPA system needs to be triggered.

4.4 Post-Market Clinical Follow-Up (PMCF) Execution

For high-risk or novel technologies, active PMCF studies must be conducted to confirm long-term clinical safety. PMCF activities may include tracking multi-center patient registries or administering structured clinical surveys to surgeons to identify rare long-term side effects or off-label use trends.

4.5 Reporting and Technical File Linkage

The output of the surveillance process must be documented in a final PMS Report. This report updates the Clinical Evaluation Report (CER), the Risk Assessment File, and the active Instructions for Use (IFU) to ensure safety information remains up to date.

5.0 Documentation Retention

All PMS Plans, literature strings, raw evaluation datasets, and final reports (PMSR/PSUR) are critical regulatory records. They must be retained for at least ten (10) years for standard devices, and fifteen (15) years for implantable options, counted from the date the last device profile was placed on the market.

6.0 Associated Reference Materials

- **SOP-QA-003:** Complaint Handling and Post-Market Intake
- **SOP-RA-005:** Vigilance and Adverse Event Reporting Architecture
- **SOP-QA-007:** Risk Management Program (ISO 14971 Integration)
- **Form-RA-004A:** Device PMS Plan Template

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], RA	Initial baseline release of formalized Post-Market Surveillance (PMS) Program Standard Operating Procedure.